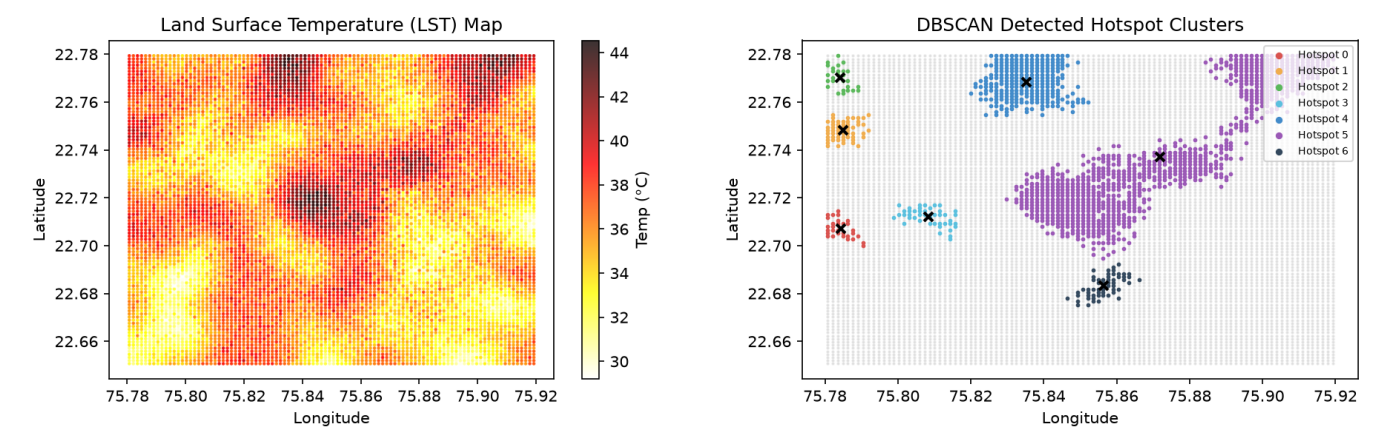


❄️ UrbanCool AI — UHI Mitigation Report

Analysis scope: Indore, MP, India | Generated: 2026-06-17 | Run: Centralized Optimization Engine

Executive Summary: This geospatial analysis report evaluates the Urban Heat Island (UHI) severity across Indore, Madhya Pradesh. Integrating Land Surface Temperature (LST) inputs with urban morphology variables and meteorological parameters, we run spatial DBSCAN clustering to isolate critical thermal hotspots. A hybrid approach combining Machine Learning (XGBoost regression + SHAP values) and a Physics-Informed Surface Energy Balance solver identifies the direct drivers of localized heating. Using a Multiple-Choice Knapsack Optimization formulation, capital is allocated to optimal mitigation scenarios within the designated budget, maximizing overall relief.

1. Geospatial Analysis & Hotspots



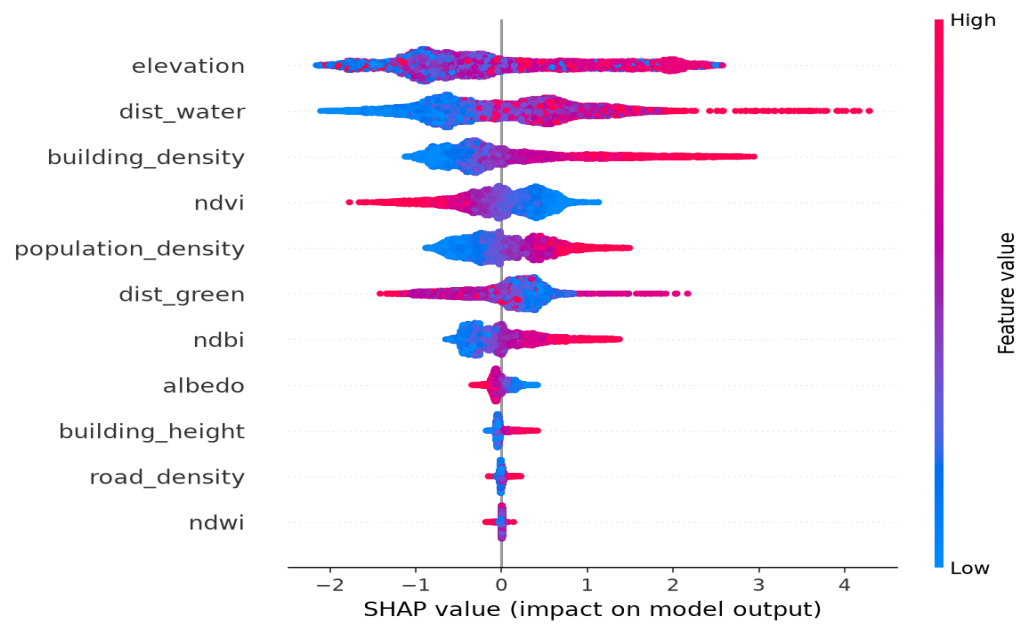
Hotspot ID	Centroid GPS	Mean Temp	Affected Pop
0	22.7071, 75.7843	39.22°C	73,662
1	22.7482, 75.7851	39.66°C	181,137
2	22.7702, 75.7841	39.01°C	94,362
3	22.7120, 75.8084	39.22°C	215,446
4	22.7683, 75.8352	40.65°C	3,033,842
...	Other hotspots summarized in dashboard	...	...

2. Machine Learning Driver Attribution & SHAP

Attribution explains **why** hotspots are hot. Green represents cooling, Red represents heating. The model's validation performance is summarized below:

ML Evaluation Metric	Value
Spatial CV R ² Score	0.0454
Spatial CV RMSE	2.300°C
Spatial CV MAE	1.794°C

Global Feature Impact on UHI Severity:



3. Physics-Informed Optimization Plan

mitigation portfolio summary: Available Budget: 50,000,000.00 INR | Spent: 48,324,561.79 INR

Hotspot ID	Recommended Intervention	Cooling Temp Drop	Cost (INR)	Pop Benefited
1	Tree Canopy	-2.50°C	17,258,772.07	181,137
4	Tree Canopy	-0.17°C	14,748,405.22	3,033,842
6	Tree Canopy	-1.35°C	16,317,384.50	355,033

Note: All cooling temperature drops are simulated using the thermodynamic energy balance solver and verified against local urban density features.